Using FP-Growth to Determine Association Rules Involving SNPs and Red-Green Color Blindness

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I. Introduction

Association rule learning refers to a series of machine learning techniques that are focused on developing probabilistic models of sets that are predictive of other sets, i.e. {A,B} => {C}. An example of this would be that if we are trying to predict a shopper's behavior, we could have an association rule that we have developed from our data suggesting that that when peanut butter and white sandwich bread are purchased, then the shopper is likely to purchase jelly ({peanut butter, sandwich bread}=> {jelly}) with a confidence of some value generated based off of our data. This field of techniques is generally considered of high interest in business-centered fields - focusing on market-basket analysis in general or even for targeted marketing towards unique customers.[1,2]

However, while it is very common to develop association rules for business transactions, the techniques within the field have seen utilization outside of business as well. One particular area of interest within biology is genetic analysis utilizing association rules. This often takes the form of seeing what are frequent sets of genes or SNPs that occur together within populations that have certain genetic disorders or are genetically predisposed to developing certain conditions - such as linking certain sets of SNPs and Crohn's disease or genotypes and bipolar disorder.[3,4] We intend to extend this same type of application to analyze whether there are certain SNP sets and association

[1] https://www.sciencedirect.com/science/article/pii/S1877050916305208
[2] https://link.springer.com/chapter/10.1007/978-3-540-78246-9_52
[3] https://ieeexplore.ieee.org/document/5170916
[4] https://pmc.ncbi.nlm.nih.gov/articles/PMC6230336/

rules within the OPN1LW or OPN1MW2 genes on the X chromosome that are responsible for, or at least highly predictive of, red-green color blindness.

II. Background - Color Blindness

Color blindness refers to the inability to fully perceive differences between certain colors or the reduced ability to perceive color in general. The most common form that this is expressed in, however, is the inability to perceive between red and green. In fact, among "European Caucasian" populations, red-green color blindness affects 8% of men. However, owing both to its recessive nature and its location on the X chromosome, these same populations only see 0.4% of women affected.[5] This is a particular trait of red-green color blindness, as both the OPN1LW and OPN1MW genes - which code for the red and green opsins in humans, respectively - are located on the X chromosome (opsins are the proteins that are responsible for detecting light in animals, determining their ability to see and, specifically, how light is perceived).[6,7]

III. Background - Single Nucleotide Polymorphisms (SNPs)

Single Nucleotide Polymorphisms (SNPs) refer to a substitution point where a single nucleotide has a different value at a certain position in the genome relative to the reference genome. For example, if the reference genome has an 'A' at a given point in the genome and an individual presents with a 'C' at that location, that is a SNP. As a SNP refers to only a single substitution, many of them ultimately have no effect on an individual, however, it has been of growing interest in the field of genetic disease

[5] https://pubmed.ncbi.nlm.nih.gov/22472762/
[6] https://www.ncbi.nlm.nih.gov/gene/5956
[7] https://www.ncbi.nlm.nih.gov/gene/2652

prediction to determine if certain SNPs are causative of specific genetic disorders or predispose individuals to certain diseases. For example, research has been done into determining the role of SNPs in predisposing individuals to diabetes, certain kinds of cancers, and many other disorders.[8,9,10] In fact, due to recent developments, researchers have become increasingly interested in utilizing SNPs along with machine learning to create disease prediction models.[11,12] In terms of color blindness, specifically, however, the National Institute of Health has determined a number of SNPs on the OPN1MW gene to have clinical significance regarding deuteranopia, green cone blindness, (rs104894914) and deuteranomaly, green cone weakness (rs104894915, rs104894916, rs724159983) - both of which lead to red-green color blindness.[13,14,15,16]

IV. Background - Association Rule Learning and FP-Growth

As stated earlier, association rule learning refers to a series of machine learning techniques where, through a set of association rules, one is able to probabilistically determine whether a grouping of items tends to lead to a result of another grouping of items. The example given before details how, given the set {peanut butter, sandwich bread}, one can assume that a shopper is likely going to also purchase the set {jelly}. While this is often used in market scenarios, it's possible to also apply to a number of other scenarios - for example, certain sets of SNPs predicting the likelihood of having a

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[8] https://www.nature.com/articles/s41598-021-86801-2
[9] https://pmc.ncbi.nlm.nih.gov/articles/PMC5746410/#s7
[10] https://ejnpn.springeropen.com/articles/10.1186/s41983-019-0093-8
[11] https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2019.00267/full#h11
[12] https://translational-medicine.biomedcentral.com/articles/10.1186/s12967-023-03939-5
[13] https://www.ncbi.nlm.nih.gov/snp/rs104894914#clinical_significance
[14] https://www.ncbi.nlm.nih.gov/snp/rs104894915#clinical_significance
[15] https://www.ncbi.nlm.nih.gov/snp/rs104894916#clinical_significance
[16] https://www.ncbi.nlm.nih.gov/snp/rs724159983#clinical_significance

disease or not. Algorithms that are used to develop association rules typically develop a number of frequent itemsets that are then used to determine association rules. The two most common algorithms for this purpose are Apriori and FP-Growth, which is used in this paper as it is generally considered to outperform Apriori and was made for that purpose.[17]

V. FP-Growth Implementation

The actual execution of the FP-Growth algorithm, which I have implemented in Python for the project, is detailed as follows:

1. We start with a set of item sets (often labelled 'transactions' owing to the background of the algorithm - being developed for market basket analysis) and a minimum support value.[18]
2. We create a frequency table of each individual item, showing how often they each show up in the input. Importantly, we do not include any items that have a frequency lower than our minimum support value.
3. Using this frequency table, we update our input set of transactions by going through each transaction, removing any transactions that are not present in the frequency table (had a frequency lower than the minimum support) and supporting the transactions in descending order based on frequency.
4. We then create a tree containing all of the transactions by adding each transaction as a sequence one by one - incrementing the attached

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[17] https://www.cs.sfu.ca/~jpei/publications/sigmod00.pdf
[18] Ibid.

frequency number of the item in the transaction if we pass through it while adding to the tree or creating a new branch and setting the starting frequencies at one if a node matching a given value in a transaction being added does not already exist.

5. After fully populating the tree, we mine the tree by first creating a table of individual items mapped to their "conditional pattern bases" which are basically just the nodes that came before them in the tree at every occurrence of the item in the tree, along with their frequencies.
6. We can use those values to create "conditional FP trees", which can be mined themselves and compared against the minimum support value to ultimately find the values that belong in a frequent itemset with the given item who has the conditional pattern base that has been turned into the tree.
7. Thus, we've managed to turn our set of transactions into a set of item sets that occur together frequently.

VI. Experiment

I utilized association rule learning via the use of FP-Growth to determine SNP sets from the OPN1LW or OPN1MW genes on the X chromosome that correlated to either red-green color blindness or normal vision.

To first obtain the SNP data, I went to OpenSNP, a website where people openly share their genomic data obtained from DNA testing services, like 23AndMe or Ancestry, as well as their self-reported phenotypic and medical data (eye color, lactose intolerance, type 2 diabetes, etc.) with the hope that enough people with similar genomic information will also share their data, and, then, researchers will utilize the available data to possibly find treatments for their conditions or simply elucidate something about their phenotype.[19] From the page for the "Colour Blindness" phenotype, I downloaded the genetic information of 12 individuals with the "Red-green colour blind" trait and that of 12 individuals with the "Not colour blind" trait. It's worth emphasizing that these traits are self-reported when considering the fact that there are people who listed their traits as being "Not color blind" or "False", as well, which both, assumedly, would equally be usable in a control group. After obtaining the genetic data of 24 total individuals, I processed their data into a single CSV where SNPs where the rows were individuals and the columns SNP headers - a given individual would have a column marked with a '1' if they had a SNP and a '0' if they did not. However, I separate columns for both red-green color blindness and normal vision as I wanted to be able to create rules that feature both of them, so simply treating the control as the absence of color blindness would not work here. The number of columns was reduced in three ways (as it would otherwise be too large to reasonably be run without significant modifications to FP-Growth and more time for the project to be run) - 1. Only SNPs on the X chromosome were considered, 2. Of those SNPs on the X chromosome, only those which were in the locations of the OPN1LW or OPN1MW genes were considered, and 3. SNPs that didn't occur at least 4 times in the dataset were removed, as they wouldn't be considered in FP-Growth, anyways, it would just slow the initial process unnecessarily.

The aforementioned CSV is loaded in as a dataframe and passed to the mlxtend library's implementation of FP-Growth (I assumed it was a simply more robust choice, as

[19] https://opensnp.org/about-us

it is a decently popular library and often used for studies centered around association rule learning). I set the minimum support value to 0.45, meaning that a value must show up in 45% of records to be considered. This value was mostly chosen to consider highly frequent items, as well as because it would train without running out of memory reliably at this value. After creating frequent itemsets using FP-Growth, I call mlxtend's association_rules function which automates the process of calculating the confidence and lift scores for the set of frequent itemsets.

VII. Results + Discussion

Running the FP-Growth and association rules script gave us two outputs - a csv full of frequent itemsets and a csv of association rules. Now, notably, there were no developed association rules or frequent itemsets that contained either the "color-blind" or "normal-vision" except for the itemsets containing only themselves in both cases. In fact, there were a number of rules developed regarding certain other SNPs predicting other SNPs but none written that suggested that certain sets of SNPs predicted either color blindness or the lack of it. Some of the values found are included below:

Antecedents	**Consequents**	**Support**	**Confidence**	**Lift**
{'i5900556'}	{'rs7053448'}	0.458	1.0	2.0
{'i4990142', 'i4990148'}	{'i5900556'}	0.458	1.0	2.182
{'rs7053448'}	{'i4990148', 'i5900556'}	0.458	0.917	2.0
{'rs7053448'}	{'i5900540'}	0.458	0.917	2.0
{'i5900540', 'rs7053448'}	{'i4990148'}	0.458	1/0	2.182

A closer inspection of the SNPs generally found SNPs from the F8 gene in the frequent itemsets and association rules.[20] As there's overlap between where the F8 gene, OPN1LW, and OPN1MW genes are found on the X chromosome, it makes sense that the filtering scheme would catch SNPs that are also located on the F8 gene, however, the overwhelming majority of the SNPs that were actually present in numbers large enough in the sample size were from the F8 gene and were highly prevalent in both the normal vision and color blind groups, which seems reasonable as the gene should not affect vision. In fact, out of the four aforementioned SNPs that are related to OPN1LW and OPN1MW, one of them was not exhibited (rs724159983) in any of the 12 color blind individuals and the other three (rs104894914, rs104894915, rs104894916) each showed once, but, interestingly the three of them all were only displayed on a single individual. That could be a sign of a pattern to look for in the future, however, it is obviously unclear from just one person.

This is also another thing of note that probably led to a lack of definitive results - the small sample size. There were only 12 people with their genomes available that I could find on openSNP that had marked 'red-green colour blind', which motivated my 12 color blind and 12 control sample size. Even if I had found any association rules involving either color blind or normal vision, it would be too small of a sample size to say anything definitive.

Finally, FP-Growth, as it turned out, was simply ill-suited to the task. There have been historical complaints online about FP-Growth's ability to handle situations where the transitions have a lot of columns, it takes massive amounts of memory to support the tree that FP-Growth develops in this situation - which is part of what motivated the minimum support value that was given.

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[20] https://www.ncbi.nlm.nih.gov/snp/?term=rs7053448

Ultimately, if I were to continue this project, I would find out how to produce significantly more data, which would probably require requesting some service and paying them for their data, and then use a different algorithm for producing these association rules - likely a modified version of FP-Growth that can run in parallel for speed-up, simply wait for Apriori, or just use a different method for determining statistical relationships between color blindness and SNPs.